MILITARY SPECIFICATION

PLOTTING BOARD, XM19, W/E

1. SCOPE

1.1 This specification covers Plotting Board: XM19 with equipment consisting of Plotting Board Case 11733754 (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-F-13926 “	Fire Control Materiel: General Specification Governing the Manufacture and Inspection of
MIL-P-14232	Parts, Equipment and Tools for Ordnance Ma- , teriel, Packaging of
MIL-I-45607	Inspection Equipment, Supply and Maintenance of

STANDARDS

Military

MIL-STD-105	Sampling Procedure and Tables for Inspection by-Attributes
M1L-STD-109	Quality Assurance Terms and Definitions

DRAWINGS

US Army, Frankford Arsenal

F11746801	Plotting Board, XM19
F11733754	Case, Plotting Board: 11733754

PACKAGING DATA SHEET

11748086	Packaging of Plotting Board XM19, W/E

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Frankford Arsenal, ATTN: SARFA-MDM, Philadelphia, Pennsylvania 19137 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 1220

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(copies of specifications, standards, drawings and Publication required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Fabrication.- The plotting board shall be manufactured in accordance with Drawing F11746801 and drawings pertaining thereto. The plotting board case shall be manufactured in accordance with Drawing 11733754 and drawings pertaining thereto.

3.2 General specification.- The contractor shall be responsible for adherence to and compliance with the following requirements of Specification MIL-F-13926:

a. Order of precedence.
b. Dimensions and tolerances.
c. Inorganic protective surface finishes.
d. Part identification and marking.
e. Workmanship

3.3 Environmental.

3.3.1 Storage temperatures.- The plotting board shall show no evidence of phvsical failure and shall meet all performance requirements of this specification when thermally stabilized at a standard ambient temperature of between 60° and 90° Fahrenheit (F) after having been exposed to, and thermally stabilized at ambient temperatures of -50° and +140°F.

3.3.2 Operating temperatures.- The plotting board shall operate freelv while exposed to and thermally stabilized at ambient temperature of -40° and +125°F, subsequent to which the plotting board shall meet the requirements of 3.4 to 3.7 inclusive when checked at a standard ambient temperature of between 60° and 90°F.

3.4 Accuracy.- With the conditions existing as designated in (a) through (c) below, and with checkpoints located on the base grid 2300 meters to the right and left of the "O" disk line and on the 2300 meter line in each quadrant (total of four points) -the board shall meet the requirements specified in 3.4.1 through 3.4.4.

a. Hole of the disk is engaged with outside diameter of pivot center of the base grid.

b. Pin bottom of the deflection arm knob is inserted in the pivot center of the base grid.

c. The "0" azimuth index line of the disk in coincidence with the "O" index line on the base grid.

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3.4.1 With the disk rotated 800 mils, the index lines on the disk shall intersect the four located points on the base grid within 10 mils.

3.4.2 The deflection arm scale, when rotated to the four established points, shall indicate 3250 meters ± 10 meters.

3.4.3 The deflection arm meter scale when superimposed on the base grid meter scales shall coincide at the 3500 meter lines within 10 meters.

3.4.4 The vernier scales of the base grid and deflection arm shall coincide within 1.0 mil.

3.5 WARPAGE

3.5.1 The maximum warp between the disk and the base of the plotting board shall not exceed 0.03 inch as measured at the outer edge of the disk.

3.5.2 The maximum warpage of the deflection arm of the plotting board shall not exceed 0.03 inch, and this warpage shall not extend in from the edges by more than 3/16 inch.

3.6 Surface finish.- The top surfaces of the disk and the base

erasures as specified on pertinent drawings. The materials shall be translucent and have no defects such as pits, digs, scratches With depth, stains, or waves on the surface that may be mistaken for pencil marks. The lines and figures on the base grid, as seen through the disk, shall be clearly legible.

3.7 Operability.- With the hole of the disk engaged with the pivot center and with the pin bottom of the deflection arm knob inserted in the pivot center, the deflection arm and disk shall be capable of rotating freely. At a standard ambient temperature of between 60° and 90°F the deflection arm and the disk shall be removable from the base with normal finger pressure and without causing permanent distortion.

3.8 Plotting Board *Case* 11733754.- The case shall conform to all requirements specified on Drawing F11733754 and drawings pertaining thereto.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.-Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth

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in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 General provisions.- The component and subassembly inspection requirements of MIL-F-13926 form a part of the Quality Assurance Provisions of this specification. Definitions of inspection terms shall be as listed in MIL-STD-109.

4.2 First article (initial production) approval.- The requirement for first article approval and the responsibility (Government or contractor) for first article testing shall be as specified in the contract. The sample for first article approval tests shall consist of three (3) plotting boards, with cases, plus three (3) each of all items covered by SQAP. The sample shall be manufactured in the same manner, using the same materials, equipment, processes, and procedures as used in regular production. All parts and materials, including packaging and packing, shall be obtained from the same source of supply as used in regular production.

4.2.1 Government testing.- When the Government is responsible for conducting first article approval tests, the contractor, prior to submitting the sample to the Government, shall inspect the sample to insure that it conforms to all the requirements of the contract and submit a record of this inspection with the sample, including certificates of conformance for materials.

4.2.2 Contractor testing.- When the contractor is responsible for conducting first article approval tests, the sample shall be inspected by the contractor for all the requirements of the contract. The sample and a record of this inspection, including certificates of conformance for materials, shall be submitted to the Government for approval. The Government reserves the right to witness the contractor’s inspection.

4.3 Inspection provisions.

4.3.1 Submission of product.- Unless otherwise specified by the Contracting Officer, inspection lot size, lot formation and presentation of lots shall be in accordance with “Submission of Product" provisions of MIL-STD-105.

4.3.2 Examination and tests.

4.3.2.1 Components and subassemblies.- All components and subassemblies shall be inspected in accordance with the inspection provisions contained in Supplementary Quality Assurance Provisions (SQAP) listed in the technical data package (TDP). In the absence of SQAP'S, the applicable Quality Assurance Provisions of MIL-F-13926 shall apply.

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4.3.2.2 Plotting board, with case.- Examination and tests related to Section 3 herein shall be performed on a class basis in accordance with MIL-STD-105 and the sampling plans specified in Tables I, II and III herein. Examination and tests for packaging, packing and marking shall be in accordance with MIL-P-14232 and Section 5 herein. The tabulated classification of defects in Tables I, II and III shall constitute the minimum inspection to be performed by the supplier after first article approval and prior to Government acceptance or rejection by item or lot.

TABLE I - CLASSIFICATION OF DEFECTS

CLASS CRITICAL: NONE MAJOR: AQL 1..0% DEFECTIVE	REQUIREMENT	TEST PROCEDURE
101. Accuracy	3.4	4.6.3
102. Warpage between disk and base	3.5.1	4,6.4
103. Warpage, deflection arm	3.5.2	4.6.5
104. Surface finish, disk and base grid	3.6	4.6.6
105. Operability	3.7	4.6.7
106. Plotting board case	3.8	4.6.8

MINOR: NONE

NOTE: The tests in Table I shall be conducted at a temperature between 60° and 90°F.

4.3.3 Acceptance and rejection.- Rejected lots shall be screened for all defective characteristics. Removal or correction of defective units and resubmittance of rejected lots shall be in accordance with "Acceptance and Rejection" as specified in MIL-STD-105.

4.4 Special sampling.

4.4.1 General.- One plotting board, with case, shall be selected at random as a special sample from each 100 produced or from each month's production, whichever occurs first. The sample shall meet the requirements and tests in Table II.

TABLE II

NO. CHARACTERISTIC	REQUIREMENT	TEST PROCEDURE
301. Fabrication	3.1	Applicable drawing - visual
302. General specification	3.2	MIL-F-13926 - visual

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NOTE: The tests in Table II shall be conducted at a temperature between +60° and +90°F.

4.4.2 Environmental.- Three plotting boards, without case, shall be selected at random as samples from each 50 produced, or from each month's production, whichever occurs first. Each sample shall meet the requirements and tests in Table III, and shall then meet the requirements and tests in Table I.

TABLE III

NO.	CHARACTERISTIC	REQUIREMENT	TEST PROCEDURE
303.	Storage temperature (-50° and +140°F)	3.3.1	4.6.1
304.	Operability (-40° and +125°F)	3.3.2	4.6.2

4.4.3 Failure of sample.- Should any one item of a special sampling fail to meet the specified test requirements, acceptance of the represented inspection lot will be suspended by the Government until necessary corrections have been made by the contractor and resubmitted items have been approved (see 4.3.3).

4.5 Inspection equipment.- Except as otherwise provided for by the contract, the contractor shall supply and maintain inspection equipment in accordance with the applicable requirements of MIL-I-45607.

4.5.1 Government furnished inspection equipment.- Where the contract provides for Government furnished test equipment, supply and maintenance of test equipment shall be in accordance with the applicable requirements specified in MIL-I-45607.

4.5.2 Contractor furnished inspection equipment.

4.5.2.1 Government design.- All inspection equipment specified by drawing number in specification or SQAP forming a part of the contract shall be supplied by the contractor in accordance with technical data included in the technical data package list.

4.5.2.2 Contractor design.- In the absence of Government inspection equipment designs, the contractor shall design and supply inspection equipment compatible with the “Test Methods and Procedures" specified in 4.6 of this specification and with the component inspection procedures specified in “Examination” and “Test Facilities” requirements of MIL-F-13926. Since tolerance of test equipment is normally considered to be within 10% of the product tolerance for which it is intended, this inherent error in the test equipment design must be considered as part of the prescribed product tolerance limit. Thus, concept, construction, materials, dimensions and tolerances

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used in the design of test equipment shall be so selected and controlled as to insure that the test equipment will reliably indicate acceptability of a product which does not exceed 90% of the prescribed tolerance limit, and permit positive rejection when non-conforming. Construction shall be such as to facilitate routine calibration of test equipment,

4.5.2.3 Inspection equipment design requirements for temperature testing.- The volume of the test chamber shall conform to the test facilities requirements of MIL-F-13926. The air temperatureat the control sensor shall be in accordance with the specified requirements within plus or minus 2.5°F. The temperature gradient across the cross-sectional area occupied by the test items shall not exceed 0.5°F per foot in any direction. The heat source of the test chamber shall be so located that radient heat, if used, will not fall directly on the test item. Thermocouples or equivalent temperature sensors utilized to determine or control the specified chamber temperature shall be centrally located within the test chamber in the supply airsteam, or *in the* return airstream, whichever provides the specified test condition at the item under test and shall be baffled or otherwise protected against radiation effects.

4.6 Test methods and procedures.

4.6.1 Storage temperature.- Testing equipment utilized in this test shall conform to the requirements of 4.5.2.3. The Plotting board shall be placed in *the* test chamber and the temperature of the chamber reduced, over a period of 4 hours plus or minus 15 minutes, from room ambient temperature to -50°F. The plotting board shall be maintained at this temperature extreme for a minimum of 2 hours. Upon completion of the 2 hour exposurre the temperature of the chamber shall be raised, over a period of 8 hours plus or minus 15 minutes, to +140°F. The plotting board shall remain at this temperature for a minimum of 2 hours. Upon completion of the 2 hours exposure the temperature of the chamber shall be reduced, over a period of 4 hours plus or minus 15 minutes, to room ambient temperature. The plotting board shall then be removed from the test chamber and subjected to a visible and tactile examination. The plotting board shall show no evidence of physical failure, or damage. The plotting board shall then be subjected to the test in 4.6.2 to determine overall compliance with the requirements of 3.3.1.

4.6.2 Operating temperature.- Testing equipment utilized in this test shall conform to the requirements of 4.5.2.3. The plotting board shall be placed in the test chamber and the temperature of the chamber reduced, over a period of 4 hours plus or minus 15 minutes, from room temperature to -40”F. The plotting board shall be maintained at this extreme temperature for a minimum of 2 hours. At completion of the 2 hour exposure, and while at -40°F, the plotting board shall be freely operable, without binding, when normal finger pressure is applied to move the deflection arm scale and disk approximately 360 degrees, in opposition to each other, about the pivot center. Upon completion of the low temperature operability test the temperature of the Chamber shall be raised over a period of 8 hours plus or minus 15 minutes, to +125°F. The plotting board shall be maintained at this extreme temperature for a minimum of 2 hours. At completion of the 2 hour exposure and while at

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+125°F the plotting board shall be subjected to the operability test specified above. Upon completion of the high temperature operability test the temperature of the chamber shall be reduced, over a period of 4 hours plus or minus 15 minutes, to room ambient temperature. The plotting board shall then be removed from the test chamber. The plotting board shall conform to the requirements of 3.3.2 at each temperature extreme. The plotting board shall then be subjected to the test in 4.6.3 through 4.6.7 inclusive to determine overall compliance to 3.3.2.

4.6.3 Accuracy.- This test shall be conducted with the plotting board assembled as specified in 3.4.a through 3.4.c The accuracy of the plotting. board shall be verified visually with the aid of a pocket type precision comparator that contains an appropriate reticle. The comparator shall have a magnification of at least 6 power (X). The accuracy of the plotting board shall conform to the requirements of 3.4.1, 3.4.2, 3.4.3 and 3.4.4.

4.6.4 Warpage, disk- Any visual evidence of warp (gap) between the disk and base of the plotting board, around the outer periphery of the disk, shall be measured with a standard thickness or feeler gage. Any warpage between the disk and the base of the plotting board shall not exceed the tolerance specified in 3.5.1.

4.6.5 Warpage, deflection arm.- Any visual evidence of warp (bending, curling, etc.) of the deflection arm, with reference to the disk, shall be measured with a standard thickness or feeler gage. The distance, or lenght, of the allowed warpage from the arm's edge shall be measured with an appropriate steel rule. Any warpage of the deflection arm shall not exceed the tolerances specified in 3.5.2.

4.6.6 Surface finish.- The top surface of the disk and base grid of the plotting board shall be capable of accepting clear and legible markings from a pencil of a hardness 2H under normal writing pressure. Upon erasure with a commercial pencil eraser there shall be no evidence of permanent marks, smudges, or smears. The pencil markings shall be applied to the four quadrants of the disk and base grid. The materials, lines and figures on the disk and base grid shall be subjected to a visual examination. The surfaces, materials, lines and figures of the disk and base grid shall conform to the requirements of 3.6.

4.6.7 Operability.- This test shall be conducted with the plotting board assembled as specified in 3.4.a through 3.4.c. When normal finger pressure is applied to move the deflection arm scale and disk approximately 360 degrees in opposition to each other, about the pivot center, there shall be no binding or interference of rotation. The deflection arm scale and disk shall be removable from base, without permanent distortion, using normal finger pressure. The plotting board shall meet the requirements of 3.7.

4.6.8 Plotting Board Case 11733754.- The case shall be subjected to a visual, tactile and dimensional inspection to determine conformance to 3.8 and Drawing F11733754.

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5. PREPARATION FOR DELIVERY

5.1 Packaging, packing, and marking.- Packaging, packing, and markine shall be in accordance with Packaging Data Sheet F11748086. The level of protection shall be as specified in the procurement document.

6. NOTES

6.1 Intended use.- The Plotting Board XM19 is used to plot data for indirect fire of the Lightweight Company Mortar, 60MM: XM224.

6.2 Ordering Data.- Procuring documents should specify the following:

a. Title, number, and date of this specification.,

b. Selection of an applicable level of preservation, packaging, and packing.

c. Applicable stock number

d. Applicable packaging data sheet number (See 5.1).

e. Provisions for first article testing.

Custodian:

Army - MU

Preparing activity:

Army - MU

Project No. 1220 - A306

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OMB Approval
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DOCUMENT IDENTIFIER AND TITLE
MIL-P-48602(MU) Plotting Board, XM19, W/E

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

   ☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

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